

Feature 1:

Chosen Values:

Material: PLA

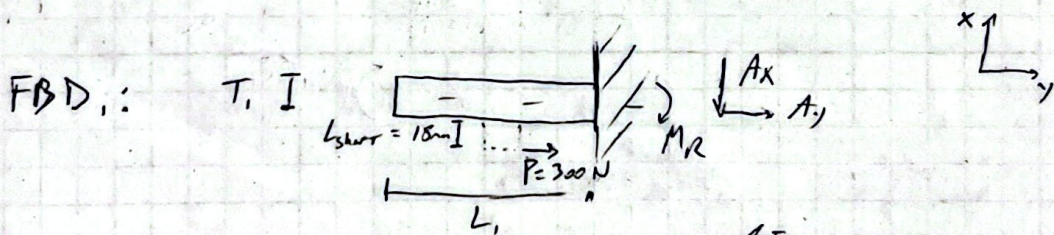
- $\sigma_{yield} = 45.2 \text{ MPa}$ (MatWeb)
- $E = 2350 \text{ MPa}$ (MatWeb)
- $L_1 = 50 \text{ mm}$
- $W_1 = 30 \text{ mm}$

KNOWN VALUES:

- Applied Load $P = 300 \text{ N}$
- Deflection $\Delta_{max} = \Delta_{max} = 0.30 \text{ mm}$
- Safety Factor (FOS) = 3

Unknown Values:

- Thickness (T_1)
- Moment (M_1)



$$\sum F_y = 0 = P - A_x$$
$$\underline{P = A_x}$$

$$\sum M = 0 = (P L_1) - M_R$$
$$\underline{M_R = P L_1}$$

$$M_R = M_1 = P L_1$$

$$M_1 = (300 \text{ N})(18 \text{ mm}) = \underline{5400 \text{ N} \cdot \text{mm}}$$

Allowable Axial Stress: (σ_{allow})

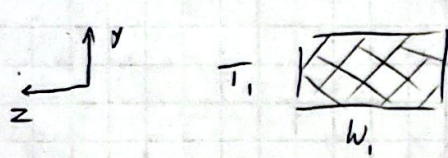
$$FOS = \frac{\sigma_{yield}}{\sigma_{allow}} \rightarrow \sigma_{allow} = \frac{\sigma_{yield}}{FOS}$$

$$\sigma_{allow} = \frac{(45.2 \text{ MPa})}{(3)}$$

$$\underline{\sigma_{allow} = 15.07 \text{ MPa}}$$

Feature 1 (cont.):

Cross-Sectional Area with Moment of Inertia (I):



$$I_1 = \frac{w_1 (T_1)^3}{12}$$

Stress Limit: Flexure Formula

$$\sigma_{allow} = \frac{M_1 (C_1)}{I_1} = \frac{M_1 (C_1)}{\left(\frac{w_1 (T_1)^3}{12} \right)} = \frac{6 M_1}{w_1 \cdot T_1^2}$$

$$T_{1,r} = \sqrt{\frac{6 M_1}{\sigma_{allow} \cdot w_1}} \quad (eq. 1)$$

$$T_{1,r} = \sqrt{\frac{6 (5400 \text{ N}\cdot\text{mm})}{(15.07 \text{ MPa}) (30 \text{ mm})}}$$

$$T_{1,r} = 8.47 \text{ mm}$$

Beam Deflection Equation:

$$\Delta L = \frac{M_1 L_1^2}{2 E I_1} = \frac{M_1 L_1^2}{2 E \left(\frac{w_1 (T_1)^3}{12} \right)}$$

$$T_{1,\Delta L} = \sqrt[3]{\frac{6 M_1 (L_1)^2}{E \cdot w_1 \cdot \Delta L}} \quad (eq. 2)$$

$$T_{1,\Delta L} = \sqrt[3]{\frac{6 (5400 \text{ N}\cdot\text{mm})}{(2350 \text{ MPa}) (30 \text{ mm}) (0.30 \text{ mm})}}$$

$$T_{1,\Delta L} = 15.65 \text{ mm}$$

$$15.65 \text{ mm} > 8.47 \text{ mm}$$

$$\boxed{\text{Governed by } 15.65 \text{ mm} = T_1}$$

Feature 2:

Chosen Values:

• Maintain width $\rightarrow W_1 = W_2 = 30\text{ mm}$

Knowns: • $\Delta L_{max} = 0.30\text{ mm}$

• $FOS = 3$

• $P = 300\text{ N}$

• $\sigma_{allow} = 15.07\text{ MPa}$

• (clearance Hole for Bolt ($d_B = 3.4\text{ mm}$))

Unknowns:

• Length ($L_{2\text{ total}}$)

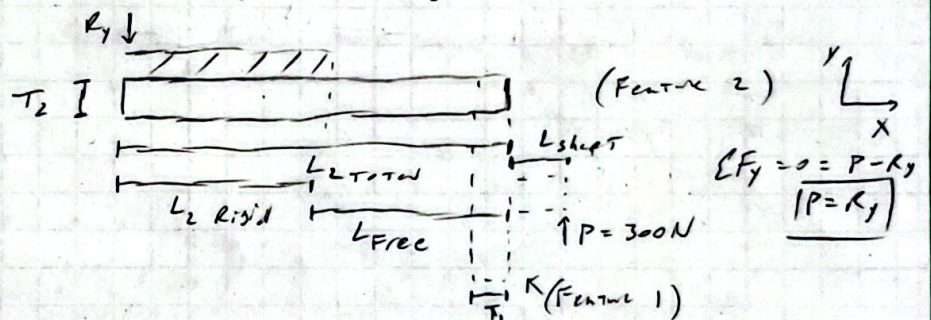
• Length (L_{rigid})

• Length (L_{free})

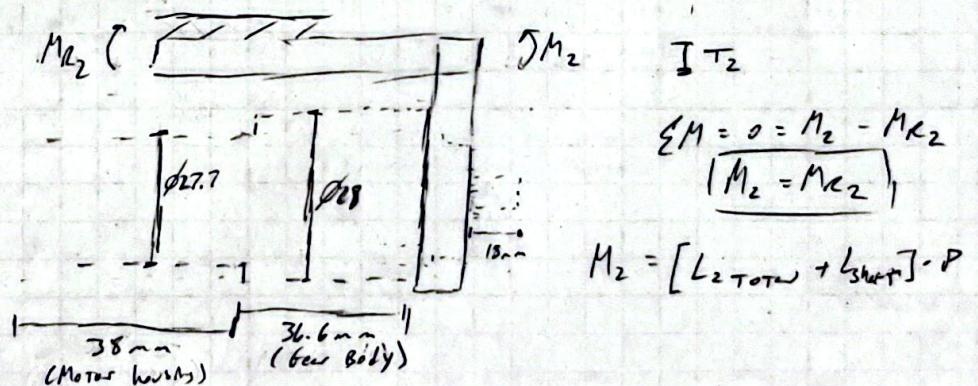
• Thickness (T_2)

FBD₂:

(1)



(2)



Determining $L_{2\text{ total}}$: $L_{2\text{ total}} = (\text{Gear Body}) + (\text{Motor Housing}) + T_1$

$$L_{2\text{ total}} = (36.6\text{ mm}) + (38\text{ mm}) + (15.65\text{ mm})$$

$$L_{2\text{ total}} = 90.25\text{ mm} \approx 91\text{ mm}$$

$$M_2 = (91\text{ mm} + 15\text{ mm}) \cdot (300\text{ N}) = 32700\text{ N}\cdot\text{mm}$$

Feature 2 (cont): Figuring L_{offset}

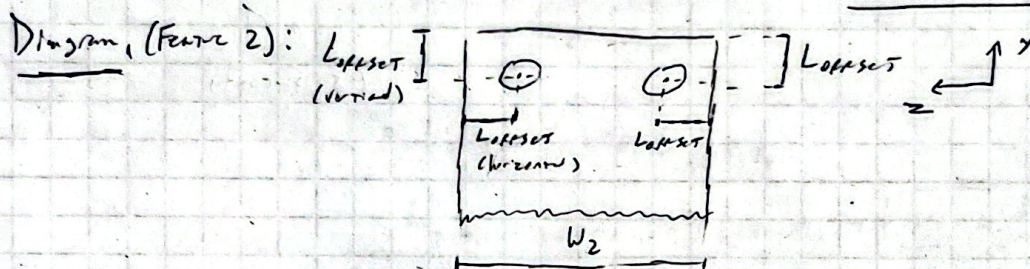
Since clearance for bolt holes were given: $\emptyset$
 $(d_B = 3.4\text{mm})$

Able to use Structural Engineering Standard:

(Distance from center of hole to closest edge = L_{offset})

$$L_{offset} = 2 \cdot d_B \quad (\text{From respective edge})$$

$$L_{offset} = 2 \cdot (3.4\text{mm}) = 6.8\text{mm} \approx 7\text{mm} = L_{offset}$$

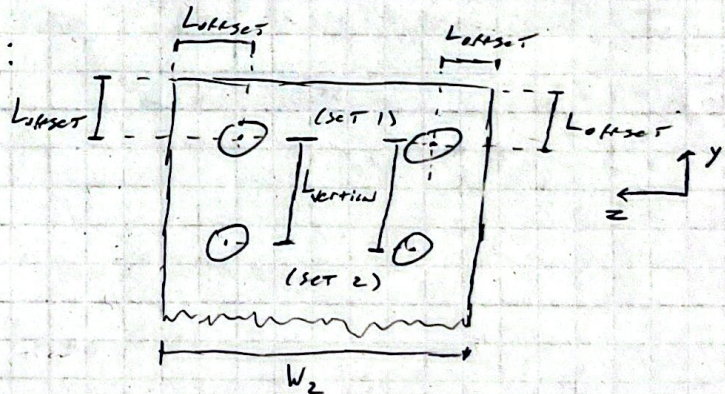


To make Holes more rigid: Apply L_{offset}

TO BOTH distances
 (Vertical & Horizontal)

Diagram₂ (Feature 2):

2-SETS of holes
 due TO Feature 2
 configuration



To Find $L_{vertical}$: Due TO Structural Engineering Standard:

$$\left(\text{Center-to-center distance From one SET TO ANOTHER} \right) = \text{"Minimum Bolt Pitch Rule"} = L_{vertical}$$

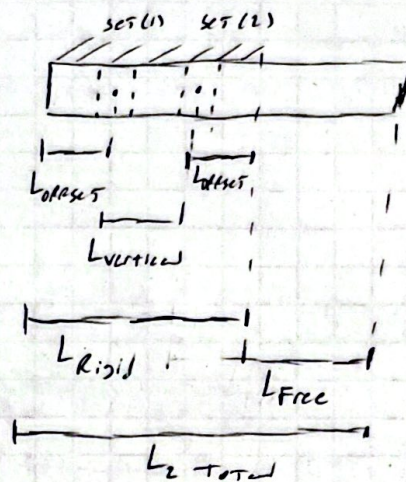
$$\text{Minimum Bolt Pitch Rule: } L_{vertical} = 4 \cdot d_B$$

$$L_{vertical} = 4(3.4\text{mm}) = 13.6\text{mm}$$

$L_{vertical} \approx 20\text{mm}$
 Bigger distance between
 SETS = more leverage TO
 resist motion

Feature 2 (L₂): Solving For L_{Rigid}

Diagram:



$$L_{Rigid} = L_{offset(2)} + L_{vertical}$$

$$L_{Rigid} = (7mm)(2) + (20mm) = \boxed{34mm}$$

Solving For L_{Free}:

$$L_{2 Total} = (\text{gear housing}) + (\text{motor body}) + T_1$$

$$L_{2 Total} = (36.6mm) + (38mm) + (15.65mm)$$

$$L_{2 Total} \approx 91mm$$

Also:

$$L_{2 Total} = L_{Rigid} + L_{Free}$$

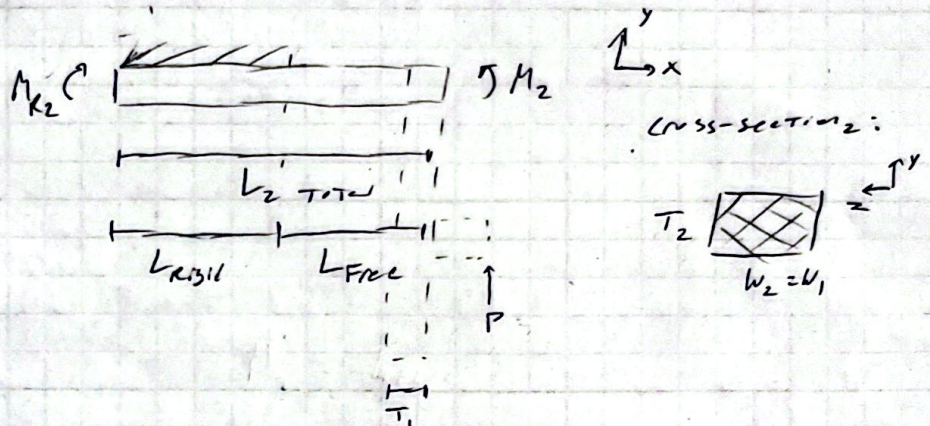
$$L_{Free} = L_{2 Total} - L_{Rigid}$$

$$L_{Free} = (91mm) - (34mm) = \boxed{57mm}$$

Feature 2 (cont.):

Now That We Solved For L_{Free}
We use This Result to calculate T_2

FBD:



T_2 Based on Flexure Formula:

$$\sigma_{allow} = \frac{M_2 (c_2)}{I_2}$$

$$\text{Moment of Inertia } I_2 = \frac{W_2 (T_2)^3}{12}$$

$$\sigma_{allow} = \frac{M_2 (c_2)}{\left(\frac{W_2 (T_2)^3}{12} \right)}$$

$$T_{2\sigma} = \sqrt{\frac{6M}{\sigma_{allow} \cdot W_2}} \quad (eq. 1)$$

$$T_{2\sigma} = \sqrt{\frac{6(32700 \text{ N}\cdot\text{mm})}{(15.07 \text{ mm})(30 \text{ mm})}}$$

$$T_{2Flex} = T_2 = 20.832 \text{ mm}$$

T_2 Based on Deflection:

$$\Delta L = \frac{M_2 (L_{Free})^2}{2 E I_2} = \frac{M_2 (L_{Free})^2}{2 E \left(\frac{W_2 (T_2)^3}{12} \right)}$$

$$(eq. 2) \quad T_{2\Delta} = \sqrt[3]{\frac{6M_2 (L_{Free})^2}{E \cdot W_2 \cdot \Delta L}}$$

$$T_{2\Delta} = \sqrt[3]{\frac{6(32700 \text{ N}\cdot\text{mm})(57 \text{ mm})^2}{(2350 \text{ MPa})(30 \text{ mm})(0.30 \text{ mm})}}$$

$$T_{2\Delta} = 31.12 \text{ mm}$$

$T_{2Flex} < T_{2\Delta} \rightarrow$ Governing Thickness

VALUES

Chosen/Given:

Material: PLA

$$\sigma_{yield} = 45.2 \text{ MPa} \quad (\text{Material})$$

$$E = 2350 \text{ MPa} \quad (\text{Material})$$

$$\Delta L_{max} = 0.30 \text{ mm}$$

$$P = 300 \text{ N}$$

$$FOS = 3$$

Feature 1	Symbolic	Value
τ_{allow}	$= \frac{\sigma_{yield}}{FOS}$	15.07 MPa
$T_{1\sigma}$	$= \sqrt{\frac{6M_1}{\tau_{allow} \cdot W_1}}$	8.47 mm
$T_{1\Delta L}$	$= \sqrt[3]{\frac{6M_1(L_1)^2}{E \cdot W_1 \cdot \Delta L}}$	15.65 mm
M_1	$= L_1(P)$	5400 Nmm
		$L_1 = 50 \text{ mm}$ $W_1 = W_2 = 30 \text{ mm}$

Feature 2	Symbolic	Value
L_{2TOTAL}	$= (\text{Gear Body}) + (\text{Motor Housing}) + T_1$ $= L_{2Rigid} + L_{2Free}$	$\approx 91 \text{ mm}$
L_{offset}	$= 2 \cdot (d_g)$	$\approx 7 \text{ mm}$
$L_{vertical}$	$= 4 \cdot (d_g)$	$\approx 20 \text{ mm}$
L_{Rigid}	$= 2(L_{offset}) + L_{vertical}$	34 mm
L_{Free}	$= L_{2TOTAL} - L_{Rigid}$	57 mm
M_2	$= P(L_{2TOTAL})$	32700 Nmm
$T_{2Flex} = T_{2\sigma}$	$= \sqrt{\frac{6M_2}{\tau_{allow} \cdot W_2}}$	20.732 mm

VALUES CONT.

Feature 2 (cont.)

	<u>Symbolic</u>	<u>VALUE</u>
T_{2DL}	$\sqrt[3]{\frac{6M_2(L_{free})^2}{E \cdot W_2 \cdot \Delta L}}$	31.12mm

Isometric Sketch:

ISO-view

